

```
1 • CREATE DATABASE company;
2 • use company;
3 • CREATE TABLE DEPT (
4     DEPTNO INT PRIMARY KEY,
5     DNAME VARCHAR(50),
6     DLOC VARCHAR(50)
7 );
8
9 • CREATE TABLE EMPLOYEE (
10     EMPNO INT PRIMARY KEY,
11     ENAME VARCHAR(50),
12     MGR_NO INT,
13     HIREDATE DATE,
14     SAL DECIMAL(10,2),
15     DEPTNO INT,
16     FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
17 );
18
19 • CREATE TABLE PROJECT (
20     PNO INT PRIMARY KEY,
21     PNAME VARCHAR(50),
22     PLOC VARCHAR(50)
23 );
24
25 • CREATE TABLE INCENTIVES (
26     EMPNO INT,
27     INCENTIVE_DATE DATE,
```

```
28      INCENTIVE_AMOUNT DECIMAL(10,2),
29      FOREIGN KEY (EMPNO) REFERENCES EMPLOYEE(EMPNO)
30  );
31
32  • CREATE TABLE ASSIGNED_TO (
33      EMPNO INT,
34      PNO INT,
35      JOB_ROLE VARCHAR(50),
36      PRIMARY KEY (EMPNO, PNO),
37      FOREIGN KEY (EMPNO) REFERENCES EMPLOYEE(EMPNO),
38      FOREIGN KEY (PNO) REFERENCES PROJECT(PNO)
39  );
40
41
42
43  • INSERT INTO DEPT VALUES
44      (10, 'HR', 'Bengaluru'),
45      (20, 'IT', 'Hyderabad'),
46      (30, 'Finance', 'Mysuru'),
47      (40, 'Marketing', 'Bengaluru'),
48      (50, 'Research', 'Hyderabad');
49
50  • INSERT INTO EMPLOYEE VALUES
51      (101, 'Arun', NULL, '2020-01-10', 50000, 10),
52      (102, 'Bhavna', 101, '2021-03-15', 60000, 20),
53      (103, 'Chitra', 101, '2019-07-20', 55000, 30),
```

```
54 (104, 'Deepak', 103, '2022-05-25', 45000, 40),
55 (105, 'Esha', 102, '2023-06-30', 48000, 50),
56 (106, 'Farhan', 101, '2024-02-10', 53000, 20);
```

57

58 • **INSERT INTO PROJECT VALUES**

```
59 (1, 'AI Research', 'Hyderabad'),
60 (2, 'Data Analytics', 'Bengaluru'),
61 (3, 'Web Portal', 'Mysuru'),
62 (4, 'Cloud Infra', 'Hyderabad'),
63 (5, 'E-Commerce', 'Bengaluru');
```

64

65 • **INSERT INTO ASSIGNED_TO VALUES**

```
66 (101, 2, 'Manager'),
67 (102, 1, 'Developer'),
68 (103, 3, 'Analyst'),
69 (104, 5, 'Tester'),
70 (105, 4, 'Support'),
71 (106, 1, 'Engineer');
```

72

73 • **INSERT INTO INCENTIVES VALUES**

```
74 (101, '2024-04-01', 5000),
75 (102, '2024-05-10', 3000),
76 (104, '2024-08-20', 4000);
```

77

Queries

1. Retrieve employee working in Bangalore, Mysuru and Hyderabad

```
78
79 • SELECT DISTINCT A.EMPNO
80 FROM ASSIGNED_TO A
81 JOIN PROJECT P ON A.PNO = P.PNO
82 WHERE P.PLOC IN ('Bengaluru', 'Hyderabad', 'Mysuru');
--
```

Output

Result Grid			 Filter Rows:	Search	Export:
	EMPNO				
☐	102				
☐	106				
☐	101				
☐	103				
☐	105				
☐	104				

2. Get employee who didn't receive incentive

```
SELECT EMPNO  
FROM EMPLOYEE  
WHERE EMPNO NOT IN (SELECT EMPNO FROM INCENTIVES);
```

Output

Result Grid		Filter Rows:	Search
EMPNO			
106			
103			
105			
NULL			

3. Get details of employee who work in same project location as dept location

SELECT

```
E.ENAME,  
E.EMPNO,  
D.DNAME,  
A.JOB_ROLE,  
D.DLOC AS DEPT_LOCATION,  
P.PLOC AS PROJECT_LOCATION
```

FROM EMPLOYEE E

JOIN DEPT D **ON** E.DEPTNO = D.DEPTNO

JOIN ASSIGNED_TO A **ON** E.EMPNO = A.EMPNO

JOIN PROJECT P **ON** A.PNO = P.PNO

WHERE D.DLOC = P.PLOC;

Output

100%

13:92

1 error found

Result Grid

Filter Rows:

Search

Export:

ENAME	EMPNO	DNAME	JOB_ROLE	DEPT_LOCATION	PROJECT_LOCATION
Arun	101	HR	Manager	Bengaluru	Bengaluru
Bhavna	102	IT	Developer	Hyderabad	Hyderabad
Farhan	106	IT	Engineer	Hyderabad	Hyderabad
Chitra	103	Finance	Analyst	Mysuru	Mysuru
Deepak	104	Marketing	Tester	Bengaluru	Bengaluru
Esha	105	Research	Support	Hyderabad	Hyderabad

```

SELECT E1.ENAME AS MANAGER_NAME, COUNT(E2.EMPNO) AS EMPLOYEE_COUNT
FROM EMPLOYEE E1
JOIN EMPLOYEE E2 ON E1.EMPNO = E2.MGR_NO
GROUP BY E1.ENAME
HAVING COUNT(E2.EMPNO) = (
    SELECT MAX(EMP_COUNT)
    FROM (
        SELECT COUNT(E2.EMPNO) AS EMP_COUNT
        FROM EMPLOYEE E1
        JOIN EMPLOYEE E2 ON E1.EMPNO = E2.MGR_NO
        GROUP BY E1.EMPNO
    ) AS TEMP
);

```

MANAGER_NAME	EMPLOYEE_COUNT	
Arun	3	

```

SELECT
    M.ENAME AS MANAGER_NAME, M.SAL AS MANAGER_SALARY
FROM
    EMPLOYEE M
WHERE
    M.EMPNO IN (SELECT
        E.MGR_NO
        FROM
            EMPLOYEE E
        WHERE
            E.MGR_NO IS NOT NULL)
    AND M.SAL > (SELECT
        AVG(E2.SAL)
        FROM
            EMPLOYEE E2
        WHERE
            E2.MGR_NO = M.EMPNO);

```

Result Grid



Filter Rows:

	MANAGER_NAME	MANAGER_SALARY	
	Chitra	55000.00	
	Bhavna	60000.00	


```

SELECT DISTINCT E.ENAME AS SECOND_LEVEL_MANAGER, D.DNAME AS DEPARTMENT
FROM EMPLOYEE E
JOIN EMPLOYEE M ON E.MGR_NO = M.EMPNO
JOIN DEPT D ON E.DEPTNO = D.DEPTNO
WHERE M.MGR_NO IS NULL;

```

100%  1:143 | 1 error found

Result Grid



Filter Rows:

Export: 

	SECOND_LEVEL_MANAGER	DEPARTMENT	
	Bhavna	IT	
☐	Farhan	IT	
	Chitra	Finance	


```
162 • SELECT E.EMPNO, E.ENAME, E.DEPTNO, D.DNAME
163 FROM EMPLOYEE E
164 JOIN EMPLOYEE M ON E.MGR_NO = M.EMPNO
165 JOIN DEPT D ON E.DEPTNO = D.DEPTNO
166 WHERE E.DEPTNO = M.DEPTNO;
```

100% 38:164 1 error found

Result Grid



Filter Rows:



Export:



EMPNO	ENAME	DEPTNO	DNAME
-------	-------	--------	-------
